EE2001 Electrical Circuits I:

Continuation to:

Circuit Variables

09 October 2023, Monday

Asst. Prof. Dr. M. Alparslan Zehir

E-mail : alparslan.zehir@marmara.edu.tr

Room : M4-228 (2nd floor)

Announcement

- **There is no laboratory lesson this week.**

Outline



Continuation to: 5. Voltage and Current

6. The Ideal Basic Circuit Element

7. Passive Sign Convention

8. Power and Energy

5 Voltage and Current

- Electric charge is **bipolar**. It can be **positive** or **negative**.
- It exists in **discrete quantities**, which are multiples of the **charge of an electron**:

$$e = 1.6022 \times 10^{-19}\, C$$

- **Separation of charge** results **in electric potential**, which is referred to as **voltage** in this course.
- **Motion of charge** results in electric current.
- The SI unit for voltage is **Volt (V)**.
- The SI unit for current is **Ampere (A)**.

Voltage and Current

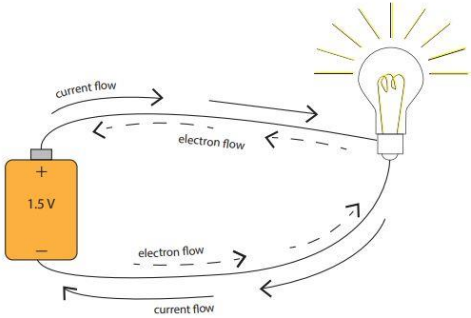
- Voltage (electric potential) is **electric potential energy per unit charge**:

$$v = \frac{dw}{dq}$$

- Electric current is **the rate of charge flow**:

$$i = \frac{dq}{dt}$$

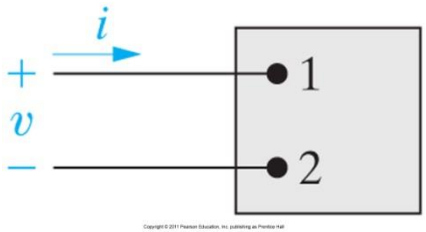
v : the voltage in volts
w : the energy in joules
q : the charge in coulombs
i : the current in amperes
t : the time in seconds



- The voltage has a **polarity** and the current has a **direction**.
- Although **in reality** electrons are moving **from negative terminal to positive terminal**, the current flow direction is **conventionally assumed from positive terminal to negative terminal**.
- Although the charge is **a discrete quantity**, there are enormous number of charge and the current is **treated as a continuous quantity**.
- If two **physically different circuit elements** have the **same current-voltage characteristics**, their circuit models are identical and **they are identical from the point of circuit analysis**.

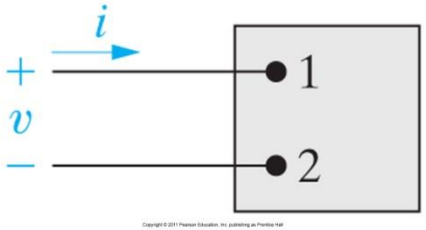
The Ideal Basic Circuit Element

- An ideal basic circuit element has **three attributes**:
  - It has only **two terminals**,
  - It is described mathematically in terms of **current and/or voltage**,
  - It **can not be subdivided** into other elements.
- Ideal basic circuit elements **form the building blocks** for constructing circuit models.



Passive Sign Convention

- You can **assume either** voltage polarity or current direction arbitrarily.
- After one of them is chosen, it dictates the **assumed** direction of **the other parameter**.
- Which means, once one of these are assigned, the corresponding equations must be written accordingly (consistency between the equations).
- These assumptions provide **reference** voltage polarity and current directions. It has nothing to do with **actual** directions.

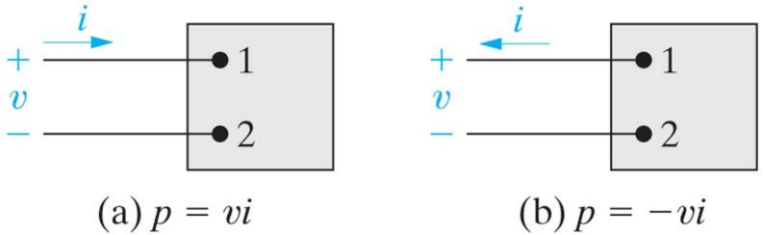


- These assumptions are done for **each element one by one**, **independent** from the other elements in the circuit.
- The most commonly used sign convention is the **passive sign convention**.

Passive Sign Convention

- **The Passive Sign Convention:** If the reference direction of the current in a circuit element is **in the direction of the reference voltage drop across the circuit element**, use a **positive sign** in any expression that relates the voltage to the current. **Otherwise**, use a negative sign.

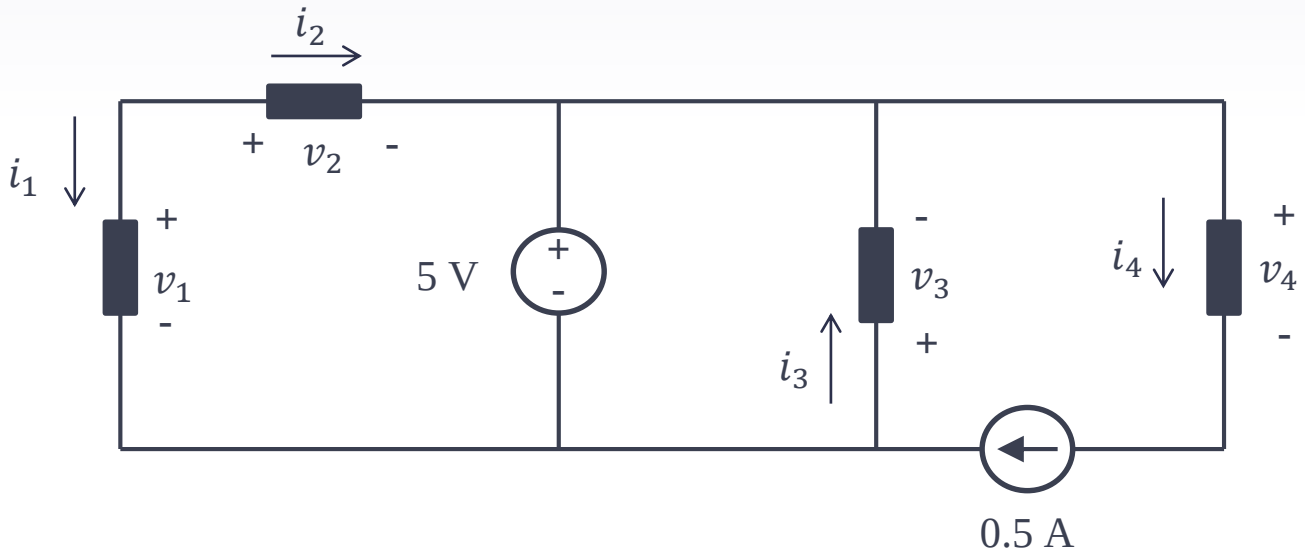
- Basically, **positive current enters the positive voltage terminal**.



Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

Passive Sign Convention

- An example:



After calculating the values for each, the negative values mean that the related parameter is in the inverse direction than your assumption (which is ok).

8 Power and Energy

- **Power** and **energy** calculations are also important in circuit analysis.
- These can be the **output** of the circuit converted to other forms.
- The circuit elements have power specifications which need to be **considered in a design process**.
- Power is **the time rate of energy consumption** or supply.
- The power can be expressed in terms of the **current and the voltage**

$$\boldsymbol{p} = \frac{dw}{dt} = \frac{dw}{dq}\frac{dq}{dt} = \boldsymbol{vi}$$

- For $\boldsymbol{p > 0}$ the circuit element **ABSORBS / CONSUMES / DEMANDS** power **from the circuit**.
- For $\boldsymbol{p < 0}$ the circuit element **DELIVERS / SUPPLIES/ PROVIDES** power **to the circuit**.

End of the topic: Circuit Variables

Any questions?

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist

Asst. Prof. Dr. M. Alparslan Zehir

E-mail : alparslan.zehir@marmara.edu.tr

Room : M4-228 (2nd floor)

EE2001 Electrical Circuits- I:

Circuit Elements

09 October 2023

From our course book*

Developing models that provide an understanding **that is imperfect but adequate** for solving practical problems **lies at the heart of engineering**.

*James W. Nilsson, Susan A. Riedel: "**Electric Circuits**", 9th Ed., Prentice Hall, New Jersey, 2011.

Outline



1. **Introduction to Circuit Elements**
2. **Voltage and Current Sources**
3. **Electrical Resistance (Ohm’s Law)**

1 Introduction to Circuit Elements

There are five ideal circuit elements:

- Voltage sources
- Current sources

} Active elements: can generate electrical energy

- Resistors
- Inductors
- Capacitors

} Passive elements: cannot generate electrical energy

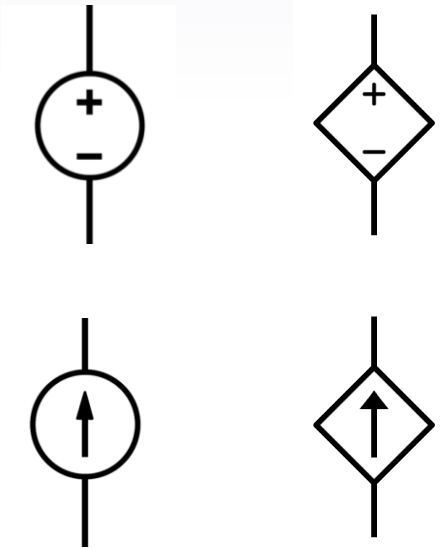
First we will learn, → **Why?**

- Voltage sources
- Current sources
- Resistors

- Many practical system can be modelled with them
- Relative simplicity, algebraic equations

2

Voltage and Current Sources



2 Voltage and Current Sources

- **An electrical source** is a device which is capable of converting non-electric (other type of) energy to electric energy and vice versa (e.g. battery, dynamo).

- They either deliver or absorb electric power, **maintaining either voltage or current**. This behavior led to the development of the **ideal voltage source** and **ideal current source** as basic circuit elements.

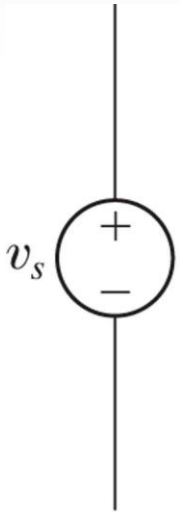
- **An ideal voltage source** maintains a specified voltage across its terminals regardless of the current flowing in those terminals.

- **An ideal current source** maintains a specified current through its terminals regardless of the voltage across those terminals.

2 Voltage and Current Sources

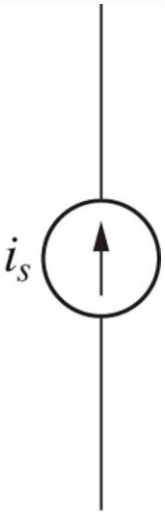
- The circuit symbols for independent sources:

Ideal independent voltage source



Value of the supplied voltage and the reference polarity

Ideal independent current source



Value of the supplied current and the reference direction

Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

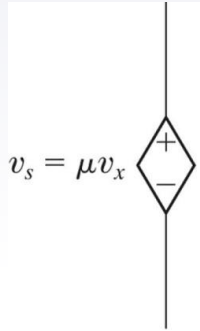
2 Voltage and Current Sources

- There are also **dependent sources.**
- The value of a dependent source depends on **the value of the current or voltage somewhere else in the circuit**.
- The value of the dependent source can be specified **only when the value of the voltage or current it depends on is known**.
- Dependent sources are also known as **controlled sources**.
- There are **four possible types** of dependent sources.

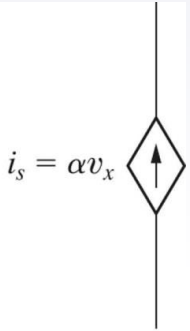
2 Voltage and Current Sources

- The circuit symbols for dependent sources:

Voltage-controlled voltage source



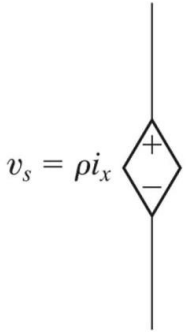
μ is a dimensionless multiplying constant



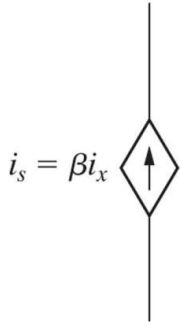
Voltage-controlled current source

α has the dimension amperes per volt

Current-controlled voltage source



ρ has the dimension volts per ampere



Current-controlled current source

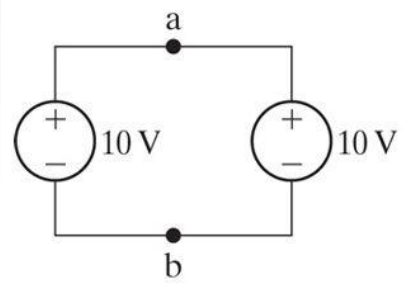
β is a dimensionless multiplying constant

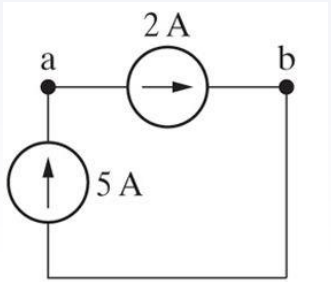
Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

2 Voltage and Current Sources

- Examples:

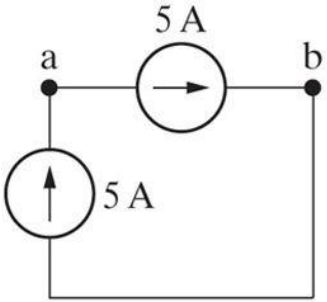
Valid

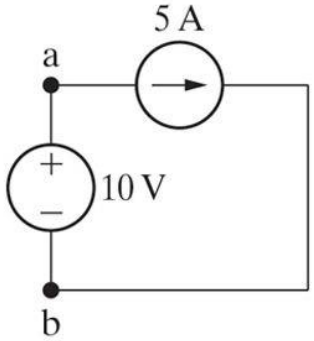




Invalid

Valid



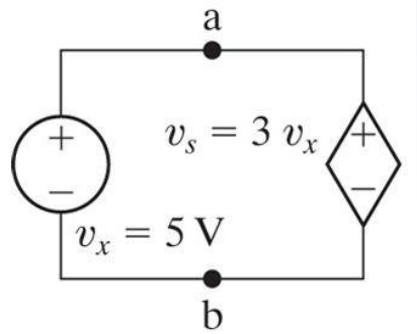


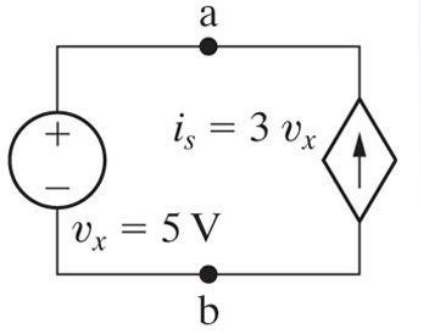
Valid

2 Voltage and Current Sources

- Examples:

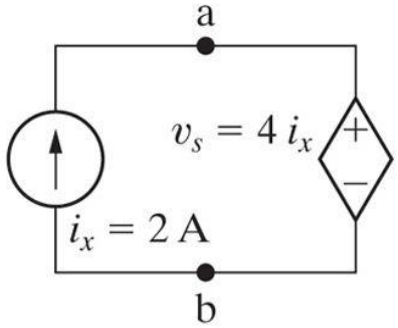
Invalid

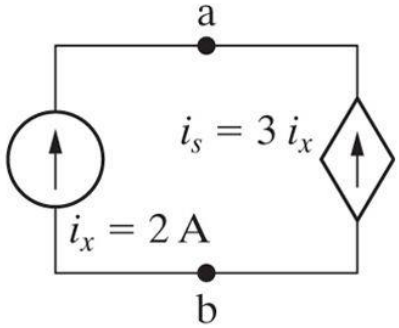




Valid

Valid

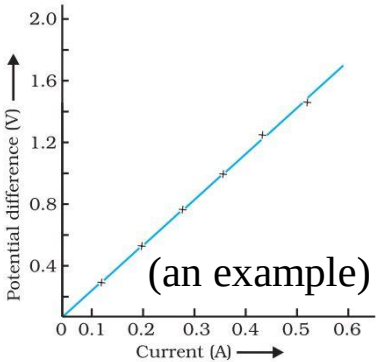




Invalid

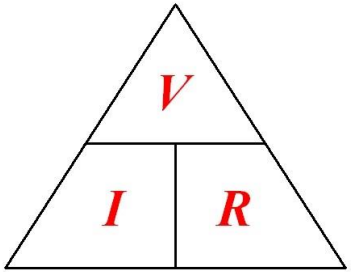
3

Electrical Resistance (Ohm's Law)



(an example)

$$V = I\,R$$



$$I = \frac{V}{R} \qquad R = \frac{V}{I}$$

3 Electrical Resistance (Ohm's Law)



Resistor symbol

- **Resistance** is the ability of materials to impede the flow of current.
- **Resistor** is the circuit element used to model this behavior.
- Moving electrons that make up electric current **interacts with and being resisted by** the atomic structure of the material. Through this interactions some amount of electrical energy is **converted into thermal energy** and **dissipated in the form of heat.**
- This **may be undesirable** in many applications. On the other hand, several electrical devices **take advantage of resistance** such as **stoves, irons, space heaters, water heaters**.
- Materials such as **copper** and **aluminum** have very small resistances (they are **good conductors**). Usually they are **not modeled as a resistor** in the circuit.

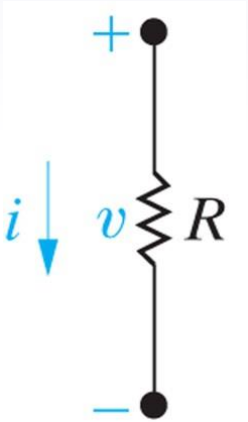
Electrical Resistance (Ohm's Law)

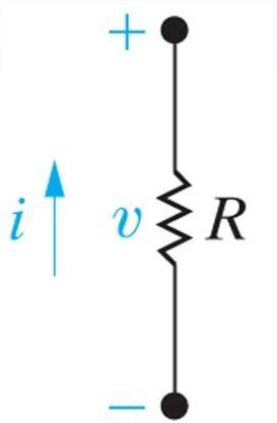


Resistor symbol with an example value

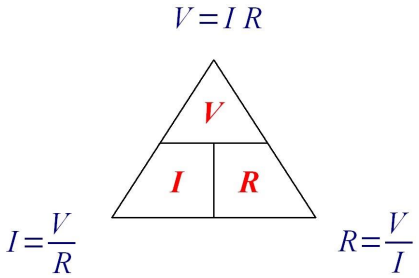
- The current-voltage characteristics of resistors obey **Ohm's Law**, using the **passive sign convention**:

$$v = Ri$$





$$v = -Ri$$



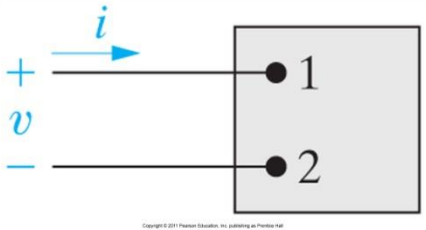
- The SI unit for resistance is **Ohm (Ω)** **(1Ω = 1V/1A)**.
- This model of ideal resistor assumes a **constant resistance**. In reality **resistance can vary** (e.g. depending on the **temperature**).

Electrical Resistance (Ohm's Law)

3

Flashback to **passive sign convention** (a topic covered in the past weeks):

- You can **assume either** voltage polarity or current direction arbitrarily.
- After one of them is chosen, it dictates the **assumed** direction of **the other parameter**.
- Which means, once one of these are assigned, the corresponding equations must be written accordingly (consistency between the equations).



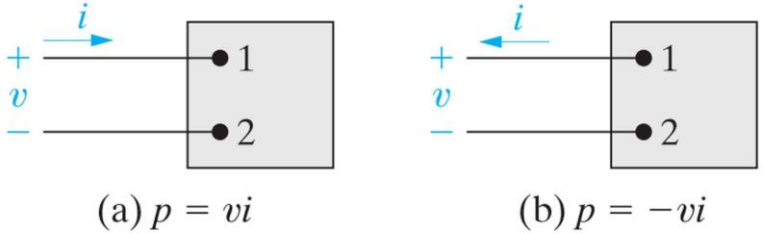
- These assumptions provide **reference** voltage polarity and current directions. It has nothing to do with **actual** directions.
- These assumptions are done for **each element one by one**, **independent** from the other elements in the circuit.
- The most commonly used sign convention is the **passive sign convention**.

Electrical Resistance (Ohm's Law)

Flashback to **passive sign convention** (a topic covered in the past weeks):

- **The Passive Sign Convention:** If the reference direction of the current in a circuit element is **in the direction of the reference voltage drop across the circuit element**, use a **positive sign** in any expression that relates the voltage to the current. **Otherwise**, use a negative sign.

- Basically, **positive current enters the positive voltage terminal**.

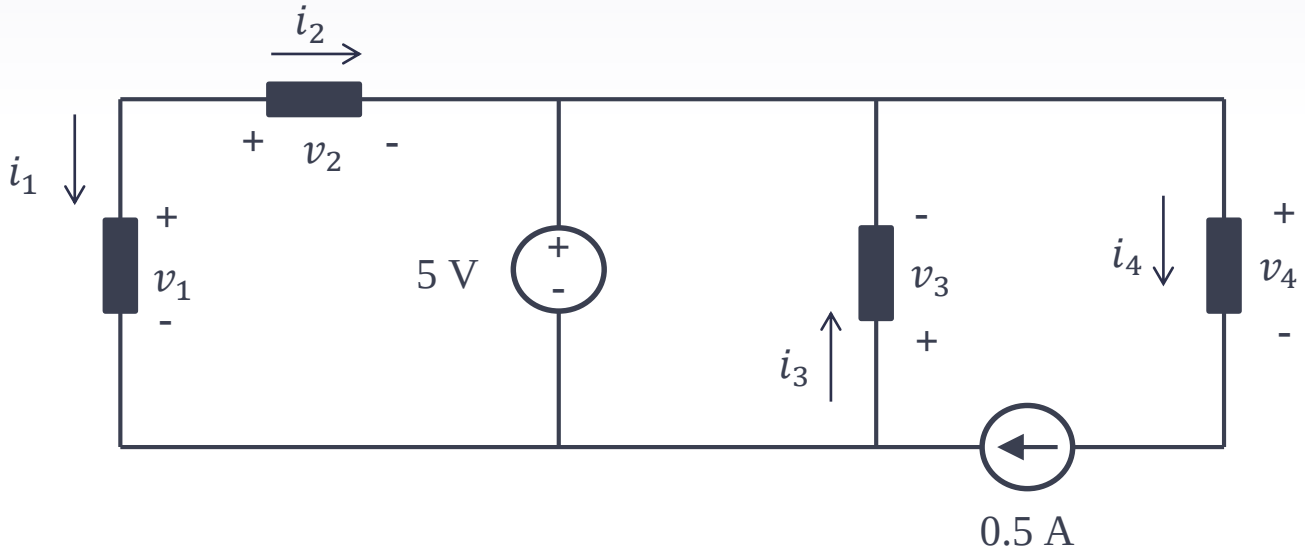


Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

3 Electrical Resistance (Ohm's Law)

Flashback to **passive sign convention** (a topic covered in the past weeks):

- An example:



After calculating the values for each, the negative values mean that the related parameter is in the inverse direction than your assumption (which is ok).

Electrical Resistance (Ohm's Law)

- The **reciprocal** of resistance is **conductance**:

$$G = \frac{1}{R}$$

- The SI unit for conductance is **Siemens (S).**
- Sometimes it is also referred to as **Mho (℧)**.

Electrical Resistance (Ohm's Law)

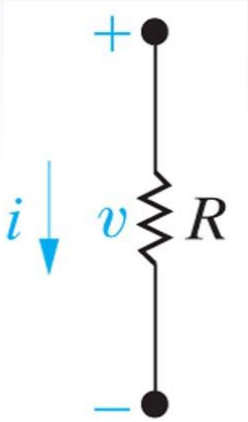
- Calculation of the power dissipated by a resistor (always positive, always absorbs power):

$$p = vi$$

$$p = (iR)i$$

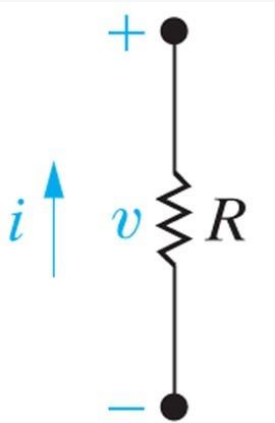
$$p = i^2 R$$

$$p = \frac{v^2}{R}$$



$$v = Ri$$

$$i = \frac{v}{R}$$



$$v = -Ri$$

$$i = -\frac{v}{R}$$

$$p = -vi$$

$$p = -(-iR)i$$

$$p = i^2 R$$

$$p = \frac{v^2}{R}$$

Electrical Resistance (Ohm's Law)

- Power equations using conductance:

$$p = (i\frac{1}{G})i$$



$$p = \frac{i^2}{G}$$

$$p = (\frac{v}{R}R)\frac{v}{R}$$

$$p = (vG\frac{1}{G})vG$$

$$p = \frac{v^2G^2}{G}$$

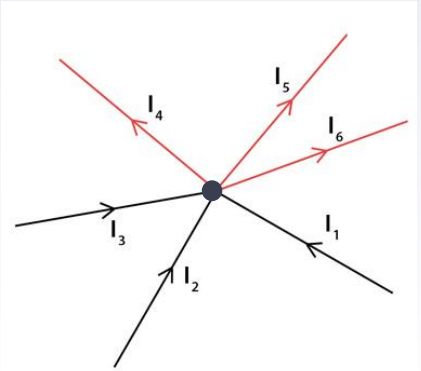
$$p = v^2G$$

4

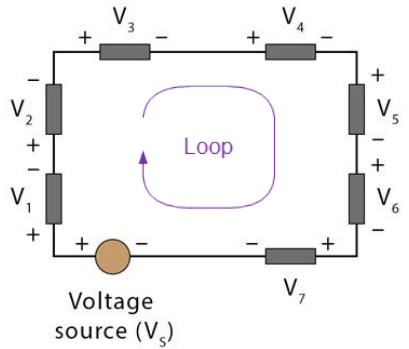
Kirchoff's Laws

Kirchoff's Current Law

Kirchoff's Voltage Law



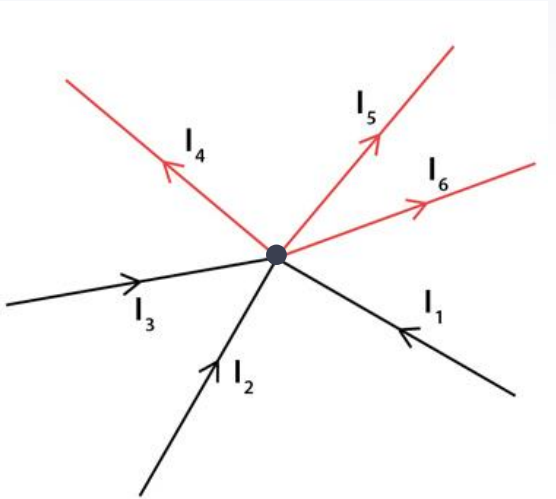
$$I_1 + I_2 + I_3 = I_4 + I_5 + I_6$$



$$V_1 + V_2 + V_3 + V_4 + V_5 + V_6 + V_7 - V_S = 0$$

4 Kirchoff's Laws

- **Kirchoff's Current Law (KCL)**: The algebraic sum of all the currents at **any node** in a circuit is **equal to zero**.
- In other words, the **sum of the currents entering a node** should be **equal to the sum the currents leaving that node** (they **can not just disappear**).
- While using **KCL** we need to **assign alebraic signs to currents**
- It possible assign a positive sign to the currents entering a node and then assign a negative sign to the currents leaving the same node.
- Alternatively, a negative sign can be assigned to currents entering a node and a positive sign can be assigned to the currents leaving that node.



$$I_1 + I_2 + I_3 - I_4 - I_5 - I_6 = 0$$

$$I_1 + I_2 + I_3 = I_4 + I_5 + I_6$$

Kirchoff's Laws

- In a circuit, a **node** is a point where two or more circuit elements **meet**.
- In the example circuit given, **a, b, c** and **d** are the nodes.
- It is necessary to **identify the nodes** in order to use **Kirchoff's Current Law**.

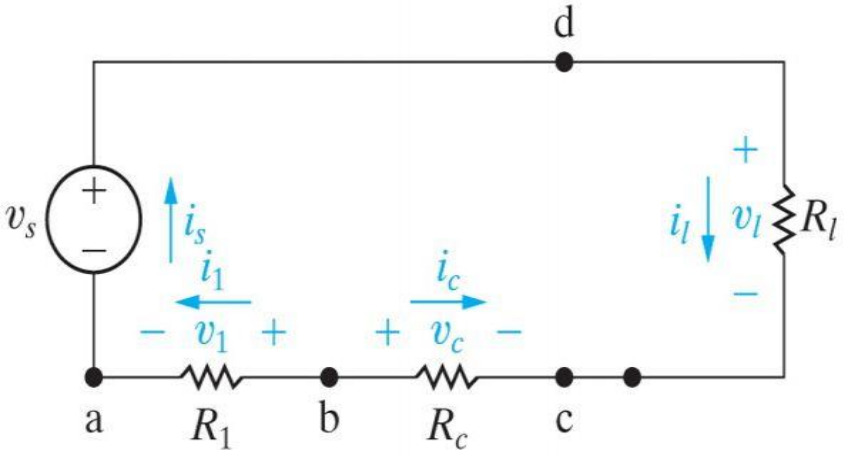
- KCL equations for the given circuit, assigning positive sign to current leaving a node and assigning a negative sign to currents entering a node:

node a $i_s - i_1 = 0$

node b $i_1 + i_c = 0$

node c $-i_c - i_l = 0$

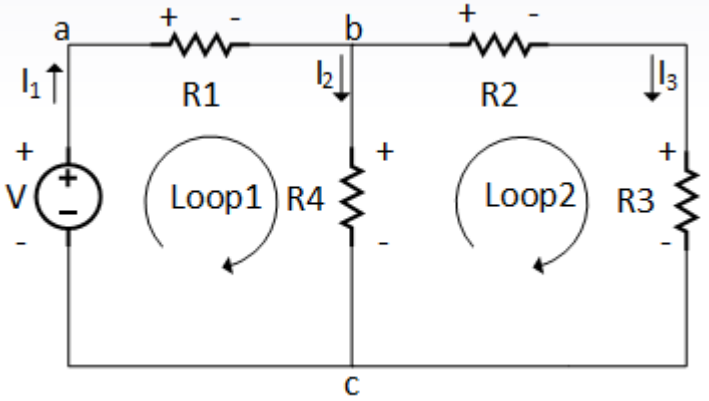
node d $i_l - i_s = 0$



4 Kirchoff's Laws

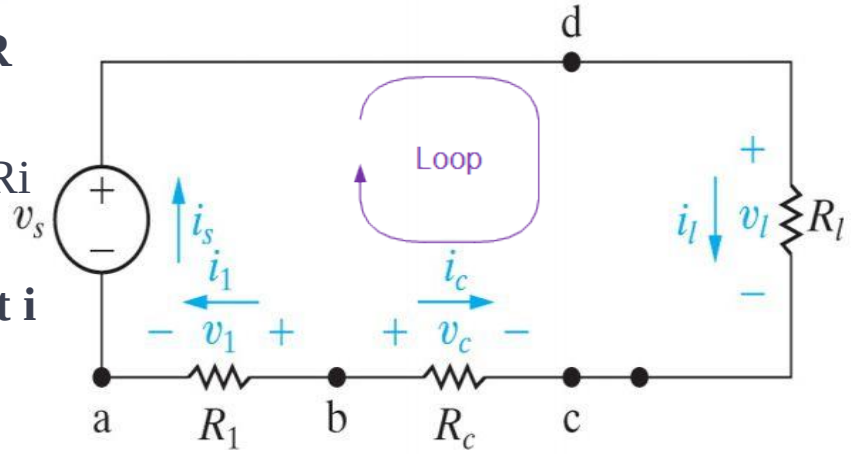
- **Kirchoff's Voltage Law (KVL)**: The algebraic sum of all the voltages around **any closed path** in a circuit is **equal to zero**.

- While using **KVL** we need to **assign algebraic signs** to voltages.



Kirchoff's Laws

- If voltage drops are used to assign positive sign, voltage rises will be considered negative.
- Alternatively, if voltage rises are used for positive sign, voltage drops will be negative.
- In this course we will mostly **use the first convention**.
- When using the first convention, as we trace the loop, **if we encounter first the positive sign** of a voltage we will write it as **positive**, **otherwise if we encounter first the negative sign** of a voltage we will write it as **negative**.
- As we trace the loop while we **go over a resistor R** if the direction of travel is **in the direction of the reference current i** we have **v =+Ri** (i.e., there is Ri voltage drop), otherwise if the direction of travel is **opposite to the direction of the reference current i** we have **v = -Ri** (i.e., there is Ri voltage rise).

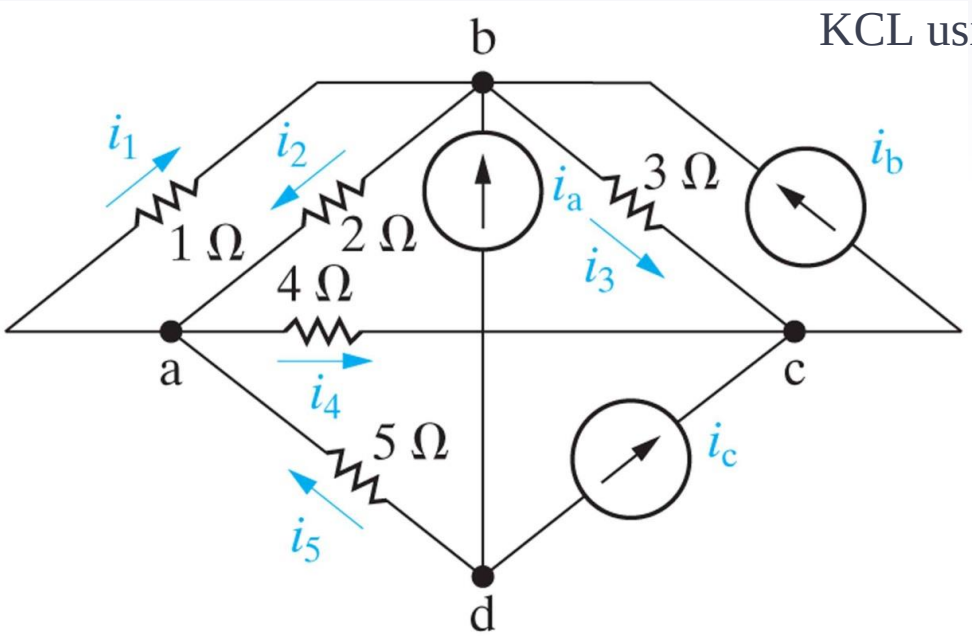


- Tracing the closed path **clockwise**, assigning **a positive sign to voltage drops**, we can write the following:

$$v_l - v_c + v_1 - v_s = 0$$

Kirchoff's Laws

- **Example 1**



Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

KCL using a positive sign for current leaving a node:

node a $i_1 - i_2 + i_4 - i_5 = 0$

node b $-i_1 + i_2 - i_a + i_3 - i_b = 0$

node c $i_b - i_3 - i_4 - i_c = 0$

node d $i_5 + i_a + i_c = 0$

End of the Lesson (09.10.23)

Any questions?

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist

Asst. Prof. Dr. M. Alparslan Zehir

E-mail : alparslan.zehir@marmara.edu.tr

Room : M4-228 (2nd floor)